Replay in the Silent Degrees of Freedom: Continual Learning Without an Offline Phase

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Abstract

Replay-based continual learning rehearses past data either in an offline phase, during which the agent stops acting, or interleaved with the live stream, where it perturbs the computation serving the current input. An agent that learns in deployment can afford neither. Motivated by local sleep, the use-dependent off-periods of individual cortical circuits in awake animals, we show that replay can instead be written into the degrees of freedom the current input leaves unused. In a network with k-winner-take-all hidden layers, confining replay updates to synapses whose presynaptic unit is silent or whose postsynaptic unit is inactive leaves the hidden computation on the current batch invariant: exactly so for silent and suppressed units, and for all but 0.3% of samples in practice. A refractory rule under which units that have just fired sit out the next competition doubles the width of this channel and carries most of the accuracy. On class-incremental split-MNIST the resulting learner, with no offline phase, matches or exceeds the best offline rehearsal schedule, matches DER++, the strongest interleaved-replay reference under backprop, and outperforms experience replay, ER-ACE and unmasked interleaved replay; in a single pass over the stream, where one offline phase per task is too few, it leads DER++ and its lead over offline rehearsal grows to 15 points. On split CIFAR-10 it again leads offline rehearsal and experience replay, but trails ER-ACE and DER++ by two to three points. The construction is not tied to the local learner: on a backprop network with k-winner-take-all hidden layers under the same schedule, refractory rotation adds half a point in five epochs and three in a single pass, and isolation again costs nothing on top of it.

1 Introduction

Foundation-model and embodied agents are increasingly expected to keep learning after deployment, from streams that are non-stationary and never task-labelled. The dominant tool against forgetting is replay from an episodic memory, and nearly every replay-based method consumes it in one of two ways: in a dedicated offline phase, a "night" during which the agent stops acting and rehearses, or by interleaving replayed samples with the live stream. Both cost an agent. An offline phase is downtime during which the system is unavailable or silently changing. Interleaved replay perturbs the computation currently serving the input.

Biology suggests a third option. Sleep-like off-periods occur *locally*, in individual cortical circuits of awake animals [Vyazovskiy et al., 2011]. Inducing such ON/OFF alternation in one hemisphere of an awake mouse discharges local sleep pressure, inducing it bilaterally during sleep deprivation rescues memory consolidation, and an equal tonic reduction of firing does not discharge the pressure [Driessen et al., 2026]. Some core functions of sleep can thus be delivered locally during wakefulness. We take this as an algorithmic motivation, not as a claim about where biological consolidation occurs,

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and ask whether the computational degrees of freedom a learner is not using can host consolidation while it acts, and what that costs.

We study the question in a deliberately constrained learner: a two-hidden-layer network with sparse population codes, trained end to end by local rules with a single forward and feedback sweep. Sparse codes expose exactly which degrees of freedom the current input does and does not use, and local updates can be confined to them synapse by synapse. The contributions are:

- **Isolated replay during inference.** A synapse may take a replay update only if its presynaptic unit is silent for the current input or its postsynaptic unit is inactive. Under k-WTA dynamics this leaves the hidden computation on the current batch exactly invariant for silent-presynaptic and suppressed units (Propositions 1 and 2), and we measure how nearly it holds elsewhere.
- **Refractory rotation.** Units that fired on the current batch sit out the next competition. This alternation, the algorithmic counterpart of the ON/OFF requirement in Driessen et al. [2026], doubles the consolidable channel and is the larger source of accuracy; random and tonic controls locate what use-dependence and all-or-none silencing each add.
- **Evaluation without an offline phase.** On class-incremental split-MNIST the system matches offline rehearsal and DER++ and beats the other interleaved-replay references, on the sequential stream and on the i.i.d. shuffle of the same data at once, at about twice the offline schedule's replay samples; on split CIFAR-10 it keeps its lead over offline rehearsal and experience replay but trails the cross-entropy backprop references. The single-pass stream, the setting closest to an agent's, is where offline rehearsal fails and the lead is largest. The same schedule on a backprop learner with k-WTA hidden layers shows that rotation and the isolation guarantee transfer beyond the local rule.

Every configuration is evaluated on two axes, the class-incremental stream and the identical learner on the i.i.d. shuffle of the same data, because consolidation during inference must not damage ordinary learning. The scale is small (MLPs on MNIST and CIFAR variants); the object of study is the mechanism and its prerequisites (Section 5).

2 Related work

Sleep-inspired consolidation. The Bazhenov line implements sleep as a global offline phase driven by noise under local plasticity [Tadros et al., 2022, Bazhenov et al., 2026, Golden et al., 2022, Kubo et al., 2025]; wake–sleep frameworks with NREM/REM stages [Sorrenti et al., 2025, Robinson et al., 2022] and systems such as SIESTA [Harun et al., 2023] likewise schedule consolidation offline, and SESLR [Lin et al., 2026] appends a noise-enhanced sleep phase to correct online learning's recency bias. None consolidates in currently unused units during inference, and the bias correction such a phase delivers is what our plastic readout receives continuously from isolated replay.

Gating and history-dependent competition. Context-dependent gating activates a subnetwork per task from task identity [Masse et al., 2018]; learned gates [Tilley et al., 2023], dendritic context [Iyer et al., 2022], task-free sparsification [Lässig et al., 2023] and recovered functional masks [McKee et al., 2026] derive the gate from the input. Refractory competition has classical precedents [Kaski and Kohonen, 1994, Maeda and Miyajima, 1999], and spiking continual learners raise firing thresholds with use [Shen et al., 2024] or randomise the top-k selection [Shen et al., 2026] to de-overlap task codes, the wake-side counterpart of our random-alternation control. Our rotation differs in purpose: recently active units are benched for one competition to enlarge a *replay channel*, not to allocate resources across tasks.

Null-space and parameter isolation; replay baselines. A mature line protects *previous* tasks: gradient or activation-subspace projections [Farajtabar et al., 2020, Saha et al., 2021, Chaudhry et al., 2019], feature-covariance null spaces [Wang et al., 2021], k-winner sparsity with such projections [Abbasi et al., 2022], frozen subnetworks [Golkar et al., 2019, Mallya and Lazebnik, 2018]. We invert the protected direction: the ongoing computation is held invariant while old-memory replay is written into degrees of freedom the current sparse support exposes as invisible. No bases or task masks are stored, and the "null space" changes with every batch at no cost. Among interleaved-replay methods, DER++ [Buzzega et al., 2020] distils stored logits and ER-ACE [Caccia et al., 2022] restricts the incoming loss to the classes present; both serve as references below. Complementary-learning-system methods [Arani et al., 2022, Pham et al., 2021] raise the value of each replayed sample but still replay at every step, and learned replay scheduling [Klasson et al., 2023] is related to the trigger study in Appendix E.

3 Method

Setting. A stream of labelled examples is presented in T contiguous tasks with disjoint class sets; no task identity is available at test time; a single shared C-way readout is evaluated on all classes after the last task (class-incremental learning). We report final accuracy A_{seq}, forgetting F, the accuracy A_{iid} of the identical learner and budget on the i.i.d. shuffle (the *static* axis), and the replay cost R in samples. The learner has an episodic memory of K raw examples with reservoir writing [Vitter, 1985]. The question is whether the offline term of consolidation can be removed entirely without losing A_{seq}, and at what price in A_{iid} and R.

Base learner. Layer activities are $a^0 = x$ and, for $\ell = 1, \dots, L$,

$$z^\ell = W^\ell a^{\ell-1} + b^\ell, \qquad a^\ell = \Phi_k\big(\sigma(z^\ell) \odot (1 - s^\ell)\big), \tag{1}$$

where σ is a rectifier, $s^\ell \in \{0,1\}^{n_\ell}$ the suppression mask of the rotation rule below, and Φ_k keeps the k largest entries of each row (k-WTA). Layers $1, \dots, L-1$ are hidden; layer L is a linear C-way readout whose pre-activation z^L is the output, scored by squared error against the one-hot target y. Errors come from one top-down sweep through learned feedback weights $B^\ell \in \mathbb{R}^{n_{\ell+1} \times n_\ell}$ (no weight transport),

$$\varepsilon^L = y - z^L, \qquad \varepsilon^\ell = \sigma'(z^\ell) \odot \Big((B^\ell)^\top \big[\kappa \tanh(\varepsilon^{\ell+1}/\kappa) \odot \mathbf{1}(a^{\ell+1} > 0)\big]\Big), \tag{2}$$

a bounded, event-gated burst signal in the spirit of predictive-coding and burst-dependent learners [Whittington and Bogacz, 2017, Payeur et al., 2021]. Forward and feedback weights follow the same local product with a shared decay λ (Kolen–Pollack alignment [Kolen and Pollack, 1994, Akrout et al., 2019]), $\Delta W^\ell \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda W^\ell$ and $\Delta B^{\ell-1} \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda B^{\ell-1}$, under Dale's law with sign-concordant feedback and 30% distance-dependent connectivity. The optimiser is heavy-ball SGD with one per-synapse velocity; masked Adam ties it (Appendix D). In development the whole constraint set cost about one point against a dense backpropagation MLP on i.i.d. MNIST.

Isolated replay. Let $X = \{x_b\}_{b=1}^m$ be the current waking batch, $a^\ell_{i,b}$ the activity of unit i on sample b, and $\mathcal{A}^\ell(X) = \{i : \exists b,\ a^\ell_{i,b} \neq 0\}$ the awake set. During the same forward step, a replay micro-batch drawn from the buffer is applied through the synapse mask

$$M^\ell_{ij} = \mathbf{1}\big[\, j \notin \mathcal{A}^{\ell-1}(X) \ \vee \ i \notin \mathcal{A}^\ell(X) \,\big], \tag{3}$$

i.e. iff the presynaptic unit is silent for the current input or the postsynaptic unit is asleep. The mask applies to the *entire* replay-induced parameter change, not only to the data gradient: with heavy-ball velocity v ($\mu = 0.9$), the local update direction $G = \varepsilon^\ell (a^{\ell-1})^\top$ and the per-step decay coefficient λ, a replay step is

$$v \leftarrow M \odot (\mu v + G) + (1 - M) \odot v, \qquad W \leftarrow W + \eta\, M \odot v - \lambda\, M \odot W, \tag{4}$$

so the velocity is frozen (not zeroed) outside the mask, the decay is confined to it, and biases follow the unit mask $\mathbf{1}[i \notin \mathcal{A}^\ell(X)]$; unmasked momentum would re-inject the update into awake synapses [McKee et al., 2026, Bricken et al., 2023]. The readout row and its bias stay plastic, so old classes remain calibrated; the guarantees below concern the hidden computation. Within one step the order is: forward pass under the current suppression, unmasked waking update, awake sets from that pass, masked replay on the updated weights; replay is inferred with the suppression removed, so a refractory unit may fire on replay and learn from it (Algorithm 1).

Proposition 1 (Pre-silent updates are exactly invisible)**.** *Fix X and let $\widehat{W}^\ell = W^\ell + \Delta^\ell$ with $\Delta^\ell_{ij} = 0$ whenever $j \in \mathcal{A}^{\ell-1}(X)$. Then every hidden activity on every sample of X is unchanged for any magnitude of Δ; if the readout row satisfies the same condition, or is held fixed, the network output is unchanged as well.*

Algorithm 1 One waking step of the default system (rotation always on, one micro-batch per batch).

Require: weights W_t, velocities v_t, suppression $s(t)$, buffer $\mathcal{M}$

$a \leftarrow \mathrm{forward}(X_t;\, W_t,\, s(t)),\ \ \varepsilon \leftarrow \mathrm{sweep}(a, y_t)$ — Eqs. (1)–(2)

$(W, v) \leftarrow$ unmasked heavy-ball step on (W_t, v_t) with $G^\ell = \varepsilon^\ell (a^{\ell-1})^\top$, rate η_w — wake update

$\mathcal{A}^\ell \leftarrow \{i : \exists b,\ a^\ell_{i,b} \neq 0\}$ for every hidden layer — from the pre-update a

$M^\ell \leftarrow$ Eq. (3), unit mask $\mathbf{1}[i \notin \mathcal{A}^\ell]$, readout row and bias unmasked

$s^\ell_i(t{+}1) \leftarrow \mathbf{1}[i \in \mathcal{A}^\ell]$ — refractory rotation

write (X_t, y_t) into $\mathcal{M}$ (reservoir)

$(X_r, y_r) \sim \mathcal{M};\ \ a_r \leftarrow \mathrm{forward}(X_r;\, W,\, s{=}0),\ \ \varepsilon_r \leftarrow \mathrm{sweep}(a_r, y_r)$ — suppression removed

$(W, v) \leftarrow$ masked step, Eq. (4), with $G^\ell = \varepsilon^\ell_r (a^{\ell-1}_r)^\top$, rate η_r, masks M^ℓ

return $W_{t+1} = W,\ v_{t+1} = v,\ s(t{+}1)$

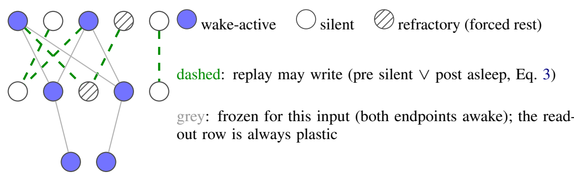

Figure 1: **Isolated replay with refractory rotation.** During each waking batch, replay from the episodic buffer is written only into hidden synapses invisible to the current input, exactly so for silent presynaptic and suppressed units; refractory rotation forces the units that just fired into the consolidable set for the next batch.

Proposition 2 (Post-asleep updates are invisible under a margin)**.** *Let Δ^ℓ and Δb^ℓ additionally move synapses and biases of units $i \notin \mathcal{A}^\ell(X)$ with active presynaptic partners, and write $\delta_{i,b} = \sum_j \Delta^\ell_{ij} a^{\ell-1}_{j,b} + \Delta b^\ell_i$. If for every such unit and every sample $\sigma(z^\ell_{i,b} + \delta_{i,b}) < \tau^\ell_{k,b}$, where $\tau^\ell_{k,b}$ is the k-th largest value entering Φ_k on sample b (strict, so a tie cannot promote the unit; when fewer than k entries are positive, $\tau^\ell_{k,b} = 0$ and the condition reads $\sigma(\cdot) = 0$), the hidden activities on X are unchanged.*

Corollary 1 (Suppressed units are exact for any magnitude)**.** *If $s^\ell_i = 1$ in (1), $a^\ell_{i,b} = 0$ for every sample and Proposition 2 holds with no margin condition.*

Proofs are in Appendix A. The propositions take the mask and the weights it protects from the same state; in the implemented order the awake sets precede the waking update, so they hold exactly for the pre-update weights and the one-step lag is part of the measured drift. The default system does not enforce the margin, so we measured it over all 16,885 isolated replay updates of the reported run (three seeds): the update altered the top hidden code of 0.33% of waking samples (0.31–0.35) and a waking prediction for 0.32%, almost all through the plastic readout (0.001% with the readout isolated), and violated the margin for $\sim 10^{-4}$ of asleep units per step. The isolation is thus exact for silent-presynaptic and suppressed units and near-exact elsewhere for the hidden code of the *current* batch. What replay does to later inputs is measured by the ablation, and Corollary 1 is why rotation makes the isolation robust.

Refractory rotation. After batch X_t, set $s^\ell_i(t{+}1) = \mathbf{1}[i \in \mathcal{A}^\ell(X_t)]$: every unit that fired sits out the next competition and is therefore asleep, and exactly consolidable, for X_{t+1}. Rotation widens the isolated channel $\rho_\ell = |\tilde{\mathcal{A}}^\ell \setminus \mathcal{A}^\ell(X)| / |\tilde{\mathcal{A}}^\ell|$, the fraction of the units $\tilde{\mathcal{A}}^\ell$ activated by the replay micro-batch that are asleep for the current input, from 0.2–0.3 to ≈ 0.53, at the price of running inference on the second-best coalition of units (Figure 1). The default system runs rotation on every batch and one replay micro-batch beside every waking batch; internal triggers that schedule sparse replay budgets, and a novelty gate on rotation, are described in Appendix E and are not needed for the results below.

Protocol. Split-MNIST (five tasks of two classes) and split CIFAR-10 (same protocol, $32{\times}32$ luminance), each with five epochs per task and, as the single-pass protocol, one epoch per task; each with its i.i.d. counterpart. Architecture d–512–256–C, k-WTA 10%, waking batches of 16, one

Table 1: **Split-MNIST, class-incremental,** $K{=}1000$ (held-out split). Sequential: five epochs per task; i.i.d.: the same learner on the shuffled stream; single pass: one epoch per task. Mean $\pm$ s.d. over six seeds.

Variant	Sequential	i.i.d.	Single pass
isolated replay + rotation (ours)	$\mathbf{91.6 \pm 0.3}$	94.1 ± 0.2	$\mathbf{91.8 \pm 0.3}$
− rotation (isolated replay on natural silence)	88.0 ± 0.7	95.4 ± 0.2	86.4 ± 1.0
− isolation (rotation, unmasked replay)	91.5 ± 0.3	93.7 ± 0.3	91.5 ± 0.8
− both (unmasked interleaved replay)	85.4 ± 3.5	93.8 ± 5.0	83.0 ± 14.9
random rotation of matched size	91.1 ± 0.7	92.3 ± 0.3	90.5 ± 0.9
replay confined to the readout	41.7 ± 10.9	81.2 ± 2.5	25.1 ± 3.0
mirror isolation (protect the past instead)	79.4 ± 3.5	90.2 ± 3.0	80.9 ± 4.1
soft rotation (gain 0.5 instead of 0)	89.9 ± 0.7	84.6 ± 0.8	87.8 ± 1.4
no replay	19.4 ± 0.0	95.1 ± 0.1	18.1 ± 0.5
offline rehearsal, same learner	89.0 ± 3.5	—	76.9 ± 3.4
offline rehearsal, narrow learner	88.5 ± 2.8	—	71.5 ± 1.9
BP + ER	88.8 ± 0.3	97.4 ± 0.1	88.6 ± 0.2
BP + DER++	91.5 ± 0.5	—	90.1 ± 0.7
BP + ER-ACE	90.2 ± 0.4	—	88.5 ± 0.3
BP + A-GEM	68.5 ± 7.0	—	46.8 ± 5.0

replay micro-batch of 16 per waking batch, buffer $K{=}1000$. Baselines: the offline "night" on the same learner (20 replay batches of 256 at $3\times$ learning rate after each waking epoch, the best offline schedule in our sweep) and on its preferred narrower core; interleaved replay for the local learner without the mask; and, under backprop with the same buffer, experience replay (BP+ER), DER++ [Buzzega et al., 2020], ER-ACE [Caccia et al., 2022] and A-GEM [Chaudhry et al., 2019], each at its best width and loss. To test the mechanism off the local learner, the same schedule also runs on a backprop MLP of the same width with k-WTA hidden layers (Table 3), with Adam's moments confined to the mask as the velocity is here and its learning rate selected on the development split. Six seeds on the reported split and three on the second, mean $\pm$ s.d. Configurations were selected on the official test splits, which served as the development set, and then frozen; every number reported in the main text re-trains the frozen configuration with a stratified tenth of the training data removed from the stream and evaluates on that tenth, which took part in no selection (during development it was part of the training stream and never evaluated on). A second tenth drawn with a different seed replicates the headline rows (Appendix H). Forgetting F is the mean drop of each earlier task's accuracy from its own end to the end of training. Evaluation runs with the suppression mask off (k-WTA only), so test order and batch size cannot affect it. Details of the sweeps and of the cost accounting are in Appendix C.

4 Results

Consolidation without an offline phase. Table 1 is the main result. With one isolated replay micro-batch beside every waking batch and no offline phase, the system reaches $91.6 \pm 0.3\%$ on split-MNIST. Offline rehearsal on the same learner, the reference that does have downtime, reaches 89.0 ± 3.5 on this split with a bimodal spread across seeds, and 91.5 ± 0.5 on the second held-out split; the narrower learner it prefers reaches 88.5 ± 2.8 and 89.3 ± 0.6. Consolidation during inference thus matches the offline reference on both splits and exceeds it on the first. Against the methods that, like ours, have no offline phase, the lead is consistent: $+2.8$ points over BP+ER (seed-paired, 95% interval $[+2.5, +3.2]$), $+6.3$ over unmasked interleaved replay on our own learner, $+1.5$ over ER-ACE and $+23$ over A-GEM, whose gradient projection protects the past at the expense of the present. DER++, which distils stored logits, is the one reference the system does not beat: the two tie at 91.5 ± 0.5 and 91.6 ± 0.3, a paired difference of 0.1 $[-0.3, +0.4]$. In a single pass over the stream the picture sharpens: the system loses nothing (91.8 ± 0.3), DER++ drops to 90.1 ± 0.7 and BP+ER holds at 88.6, unmasked replay becomes unstable (83.0 ± 14.9), and offline rehearsal falls to 76.9 ± 3.4 because one night per task is too few. The price on the static axis is 1.0 point against the same learner without replay (95.1) and 3.3 against dense backprop.

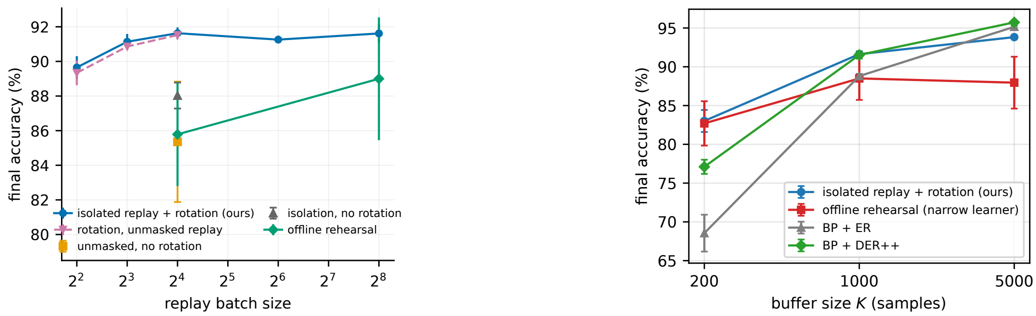

Figure 2: **Left (replay batch size):** split-MNIST accuracy against the replay batch size, the number of updates fixed within each series (one per waking batch; 20 per epoch for the night, whose samples therefore fall with its batch): the full system is flat down to 8 samples and loses 1.9 at 4 (12.7 at 2, Appendix F); rotation with unmasked replay (dashed) trails it by 0.1 to 0.3, within seed noise, and fails severely in half the seeds at batch 2; without rotation, unmasked and isolated replay and the offline night all degrade. **Right (buffer size):** accuracy against the buffer size K for the system and the replay references (held-out split, six seeds).

What rotation and isolation each buy. Removing rotation costs 3.6 sequential points (88.0 ± 0.7, the same mask on natural silence alone) and removing both rotation and isolation 6.2, with a twelve-fold larger seed deviation. Removing isolation alone costs nothing measurable in mean accuracy: rotation with unmasked replay reaches 91.5 ± 0.3, a seed-paired margin of 0.1 $[-0.3, +0.5]$, and 0.4 points on the static axis. The factors are therefore strongly sub-additive, rotation being the larger. Figure 2 (left) varies the replay batch size with the number of updates fixed. The full system is flat from 256 down to 8 samples ($91.6 \to 91.1$) and loses 1.9 at 4, where unmasked replay without rotation (85.4 ± 3.5 at 16) and the offline night (85.8 ± 3.0 at 16) have already degraded. Rotation with unmasked replay tracks the full system to within seed noise down to batch 4. At two-sample batches both lose heavily, but isolated replay degrades gracefully (79.0 ± 3.2, every seed within 75–84) where unmasked replay fails severely, below 60%, in three of six seeds (Appendix F). In the seeds and configurations tested, isolation's contribution to accuracy is the absence of those failures; what it buys at every batch size is the guarantee on the live computation, without which the outcome rests on the rotation's tolerance of the perturbation.

Controls. Replay confined to the readout row falls to 41.7 ± 10.9 and costs 13 static points, so hidden-layer replay updates are essential. A random suppression of matched size recovers most of rotation's gain (91.1 ± 0.7) but trails it by 0.5 sequential and 1.8 static points with $1.3\times$ the forgetting: alternating the coalition carries most of the effect, and resting the recent winners the rest. Soft rotation, halving rather than silencing the recent winners, loses on both axes ($89.9/84.6$), so all-or-none OFF periods beat turned-down gain, as in the biological tonic control. Mirror isolation, which bars the waking update from the synapses that served the last replay batch and leaves replay unmasked, reaches 79.4 ± 3.5; this reversed mask is one implementation of the protect-the-past direction and loses twelve points, which locates the value in the direction of protection and not in masking as such. The unmasked control is not handicapped by its replay step: over replay gains $\{1/16, 1/4, 1, 3, 10\}$ it plateaus from gain 1 to 3 (85.7, 85.4) and never closes the gap (Appendix D).

Transfer to CIFAR-10. On raw split CIFAR-10 (Table 2) the ordering among the local-learner variants and BP+ER transfers and widens: isolated replay with rotation 28.4 ± 0.8, offline rehearsal 25.1 ± 2.7, BP+ER 24.6 ± 0.6, and unmasked interleaved replay 14.6 ± 4.6, no better than no replay at all (16.3); the second held-out split gives $29.7 > 26.5 > 25.8 > 15.5$. The two cross-entropy references under backprop lead here: ER-ACE by 3.2 points and DER++ by 2.2 (seed-paired, $[+2.1, +4.4]$ and $[+1.1, +3.2]$), and in a single pass ER-ACE keeps its lead (29.5 against 26.0) while DER++ falls behind (24.4 ± 1.2). All methods are far from the i.i.d. ceiling on this input, so CIFAR-10 tests orderings rather than levels, and two survive it: removing the offline phase costs nothing against offline rehearsal, and against the best backprop references it is the local learner itself, not the consolidation mechanism, that falls short.

Table 2: **Raw split CIFAR-10, class-incremental,** $K{=}1000$. Sequential accuracy on the reported held-out split (six seeds), on a second held-out split (three seeds), and in a single pass (six seeds).

Method	Sequential	Second split	Single pass
isolated replay + rotation (ours)	28.4 ± 0.8	29.7 ± 0.9	26.0 ± 1.9
offline rehearsal, same learner	25.1 ± 2.7	26.5 ± 0.4	21.6 ± 1.0
unmasked interleaved replay	14.6 ± 4.6	15.5 ± 4.5	14.5 ± 3.9
no replay	16.3 ± 0.1	16.6 ± 0.0	11.7 ± 1.1
BP + ER	24.6 ± 0.6	25.8 ± 0.8	25.8 ± 0.7
BP + DER++	30.7 ± 0.8	31.0 ± 0.6	24.4 ± 1.2
BP + ER-ACE	$\mathbf{31.6 \pm 1.0}$	$\mathbf{31.7 \pm 0.8}$	$\mathbf{29.5 \pm 0.8}$
BP + A-GEM	16.5 ± 0.2	16.7 ± 0.2	15.3 ± 0.3

Table 3: **The same schedule on a backprop learner** (512–256, k-WTA 10%, Adam, waking batches of 16 with one 16-sample replay micro-batch each, $K{=}1000$; held-out split, six seeds; learning rate selected on the development split).

BP + k-WTA	Sequential	i.i.d.	Single pass
interleaved replay (unmasked)	91.7 ± 0.4	97.2 ± 0.1	89.3 ± 0.6
isolated replay	84.7 ± 2.6	97.6 ± 0.1	81.9 ± 2.1
interleaved replay + rotation	$\mathbf{92.2 \pm 0.3}$	96.7 ± 0.3	92.7 ± 0.3
isolated replay + rotation	92.1 ± 0.3	96.6 ± 0.1	$\mathbf{92.8 \pm 0.2}$

Cost. Samples are not the only cost. The headline schedule replays $2.2\times$ the night's samples (2.7×10^5 against 1.2×10^5) in $35\times$ as many updates (16,885 masked micro-batch updates against 480 night batches). Two schedules meet the night's budget within $1.1\times$ and keep the accuracy: eight-sample micro-batches at every waking batch (91.1 ± 0.4) and 16-sample ones at every second batch (91.2 ± 0.2). Accuracy therefore depends on the frequency of small isolated updates rather than on the number of replayed samples. In wall-clock on one CPU thread the headline run takes 8.8 minutes against 1.4 for the night, 8.6 for unmasked interleaved replay and 1.0 with no replay, in an unoptimised implementation whose mask construction is dense: the per-batch replay step, not the sample count, is the online price of consolidating during the stream (Appendix C).

Buffer size. Figure 2 (right) varies the buffer. At $K{=}200$ the system leads every reference (83.0 ± 1.4 against 82.7 ± 2.9 for offline rehearsal, 77.1 ± 0.9 for DER++ and 68.5 ± 2.4 for BP+ER); at $K{=}1000$ it ties DER++; at $K{=}5000$ the backprop references pull ahead (95.7 for DER++ and 95.2 for BP+ER against 93.8 ± 0.2), while offline rehearsal on the local learner saturates near 88. The mechanism's advantage is therefore largest where an agent's memory is smallest, and at large buffers the ceiling is the local learner's own i.i.d. accuracy (94.1), not the mechanism.

The mechanism on a backprop learner. Nothing in the construction requires the local rule, so Table 3 runs the same schedule on a backprop MLP of the same width with k-WTA hidden layers at the same 10% fraction. Rotation transfers: it lifts interleaved replay from 91.7 ± 0.4 to 92.2 ± 0.3 (seed-paired $+0.5$ $[+0.2, +0.8]$) and from 89.3 to 92.7 in a single pass ($+3.4$ $[+2.8, +4.0]$), to the level of the local learner and of DER++ under its own batch-256 protocol (91.5), at a static price of 0.5–0.9 points. Isolation on top of rotation again costs nothing (-0.1 $[-0.3, +0.1]$) and brings the guarantee. Isolation alone, however, hurts the backprop learner (84.7 ± 2.6, seven points below interleaved replay) where it helped the local one: without rotation the channel that natural silence leaves open is too narrow for this learner, so on backprop the guarantee is affordable only through the rotation. The schedule itself is also worth three points to backprop (interleaved replay in 16-sample micro-batches reaches 91.7 against 88.8 for BP+ER in batches of 256).

5 Discussion

What this offers agents. The construction is a recipe with two prerequisites, both of which can be checked on a given architecture. The representation must be sparse enough that a batch leaves most hidden units silent, and the replay update must be maskable synapse by synapse. Nothing in Propositions 1 and 2 depends on how the update is computed; we studied a local learner because its sparse codes and local rules make both prerequisites hold by construction, and Table 3 shows the same schedule on a backprop learner with k-WTA layers: rotation transfers, isolation costs nothing on top of it, and isolation alone does not transfer. Top-k routed mixture-of-experts layers expose, per token, the experts a token leaves silent; whether the routes of a whole batch leave enough experts idle is a measurement, since load balancing pushes against it. Under these prerequisites, consolidation can run in the silent degrees of freedom between the steps of the stream, with no separate offline phase, under a conditional guarantee on the computation in use, at a higher total compute ($6\times$ the night's wall-clock, unoptimised). We treat the biology as analogy, not as mechanistic equivalence; the refractory rule mirrors the ON/OFF finding of Driessen et al. [2026], and the dissociation of triggers in Appendix E is a testable prediction.

Limitations. All results are at MLP scale on MNIST and CIFAR-10, with six seeds on held-out training data and three on a second held-out split. The lead over the best offline night is 2.6 points on the first split and 0.3 on the second, where the night's seed variance vanished; the leads over the interleaved-replay references hold on both. DER++ ties the system on split-MNIST and trails it only in the single pass, and on raw CIFAR-10 ER-ACE and DER++ lead it by two to three points, so the comparison with interleaved replay is not a claim of a new accuracy level; the contribution is consolidation with the live computation held invariant, at no loss on split-MNIST against the strongest such reference. Rotation keeps a static price of 1–2 points at this activity fraction. The rule's synaptic decay is tuned once per dataset; a drive-referenced controller that replaces it by one ratio target and, with locally learned skip synapses, holds a five-hidden-layer network at the two-layer level is reported in Appendix G. Code can be found at `https://anonymous.4open.science/r/IaC-CL-with-no-offline`

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A Proofs

Proof of Proposition 1. By induction on ℓ, for every sample b. Assume $a^{\ell-1}_{\cdot,b}$ is unchanged (true for $\ell = 1$ since $a^0_{\cdot,b} = x_b$). For any unit i, $\tilde{z}^\ell_{i,b} - z^\ell_{i,b} = \sum_j \Delta^\ell_{ij} a^{\ell-1}_{j,b}$, and each term vanishes: if $j \in \mathcal{A}^{\ell-1}(X)$ then $\Delta^\ell_{ij} = 0$, otherwise $a^{\ell-1}_{j,b} = 0$ for every b. Hence $z^\ell_{\cdot,b}$, and $a^\ell_{\cdot,b}$ through (1), are unchanged; the induction closes at the last hidden layer, and at the output layer whenever the readout row satisfies the same condition. □

Proof of Proposition 2. A unit outside $\mathcal{A}^\ell(X)$ contributes $a^\ell_{i,b} = 0$ on every sample. After the update its pre-activation on sample b is $z^\ell_{i,b} + \delta_{i,b}$; under the stated margin it is still excluded by Φ_k (or rectified to zero), so $a^\ell_{i,b} = 0$ still. The sample's original winners keep their pre-activations, because their synapses from active presynaptic units lie outside Δ^ℓ by the mask and their silent-presynaptic terms vanish as in Proposition 1; so the same k units win with the same values, every other unit's input is untouched, and the layer's activities are identical. When fewer than k units are positive on sample b, $\tau^\ell_{k,b} = 0$ and the condition reads $\sigma(z^\ell_{i,b} + \delta_{i,b}) = 0$: a zero-valued entry carries no activity whether or not Φ_k selects it. The readout row is excluded from both propositions in the default system because it is deliberately kept plastic. □

B One training step and the three channels

Algorithm 1 lists one waking step of the default system with the state each quantity is read from. Two conventions matter for the guarantees. First, the awake sets are read from the waking forward pass, i.e. from the weights before the unmasked waking update, while the masked replay step acts on the updated weights; Propositions 1 and 2 hold verbatim for a mask computed on the weights being updated, and the drift reported in Section 3 is measured under the implemented order (re-inferring the batch after the waking update and masking on the post-update awake sets lowers the hidden-code rate from 0.32% to 0.30% in a development run, so the residual drift comes from the unenforced margin, not from the ordering). Second, the replay micro-batch is inferred with the suppression removed, so a refractory unit may fire on replay and receive the update (Corollary 1 makes that update invisible to the waking batch for any magnitude). The waking step is $\eta_w = \eta\, m/256$ for a waking batch of m samples (the per-epoch drift of a batch-256 protocol), the replay step $\eta_r = 3\eta$ per micro-batch, with $\eta = 0.02$ throughout.

The three channels. Table 4 lists the three ways a synapse enters the mask (3) and what each guarantees; the rotation of Section 3 exists to move synapses from the second row into the third.

Channel	Condition	Invisible to X	Source
presynaptic silent	$j \notin \mathcal{A}^{\ell-1}(X)$	for any magnitude	Prop. 1
postsynaptic naturally asleep, active pre	$i \notin \mathcal{A}^\ell(X)$, $s^\ell_i = 0$	under the margin $\tau^\ell_{k,b}$	Prop. 2
postsynaptic suppressed	$s^\ell_i = 1$	for any magnitude	Cor. 1

Table 4: **The three consolidation channels** opened by the synapse mask.

A numerical example. On one sample with $k = 2$ and $\sigma(z) = (3, 2, 0.5, 0)$, units 3 and 4 are asleep and $\tau_k = 2$. Any replay update with $\sigma(z_3 + \delta_3) < 2$ and $\sigma(z_4 + \delta_4) < 2$ leaves the activity vector $(3, 2, 0, 0)$ unchanged, because the two winners' own inputs are not touched; if unit 4 is refractory ($s_4 = 1$) its bound is void and δ_4 may be arbitrarily large; and every synapse leaving unit 4 into the next layer may move by any amount, since its presynaptic activity is zero.

C Protocol details, baseline tuning and cost accounting

Split CIFAR-10 uses 32×32 luminance so that distance-dependent wiring keeps its image geometry. The local learner uses 80% excitatory units and 30% distance-dependent connectivity; the buffer uses reservoir writing; the narrow night reference uses a 256–128 core, its preferred width in our sweeps. Runs are deterministic at a fixed CPU thread count.

Baseline tuning. Selection everywhere was sequential accuracy at $K{=}1000$ on the official test split (the development set), with the static axis reported alongside; the comparisons reported are on the held-out split (Appendix H). The offline-night reference was swept over replay batch count (20, 40 per night), replay batch size (16, 64, 256), learning-rate gain (3×), core width (256–128 and 512–256), buffer size (200, 1000, 5000) and timing (clocked after each epoch, novelty-timed, surprise-timed), and its best sequential cell is reported. BP+ER was swept over learning rate ($3{\cdot}10^{-4}$, 10^{-3}, $3{\cdot}10^{-3}$), width and buffer size. DER++, ER-ACE and A-GEM share BP+ER's network, Adam optimiser, batch of 256 waking samples with a replay batch of the same size, and reservoir buffer; each was swept over width (256–128, 512–256) and, where the method allows it, over the loss on the incoming batch (squared error on the one-hot target, as for BP+ER, or cross-entropy), DER++ additionally over $\alpha \in \{0.01, 0.03, 0.1, 0.3, 1\}$ and $\beta \in \{0.5, 1\}$, where α multiplies the per-sample sum of squared logit differences (ten times the mean-reduced form of the published code); ER-ACE is defined with cross-entropy. The selected cells on split-MNIST are DER++ with cross-entropy, $\alpha = 0.03$ (0.3 in the mean-reduced form), $\beta = 1$ at width 512–256 (development split 92.2, with every α from 0.01 to 1 within 0.2 of it; its squared-error variants reach 89.9 at best), ER-ACE at 256–128 (90.4) and A-GEM with squared error at 512–256 (73.6; with cross-entropy A-GEM is unstable, 37–43). On raw CIFAR-10 the same sweep selects DER++ with cross-entropy, $\alpha = 1$, $\beta = 1$ at 512–256 (31.1), ER-ACE at 512–256 (31.7) and A-GEM at 512–256 (16.5, the no-replay level). The backprop k-WTA learner of Table 3 was swept over Adam learning rates $\{10^{-5}, 3{\cdot}10^{-5}, 10^{-4}, 3{\cdot}10^{-4}, 10^{-3}, 3{\cdot}10^{-3}\}$ under the local schedule; interleaved replay selects $3{\cdot}10^{-5}$ (development split 92.4) and the three other variants 10^{-4} (92.9 and 92.9 for the two rotation variants, 84.3 for isolation alone). The local learner was swept over η (a tenfold range, with a plateau over 0.01–0.05), replay batch size, cadence and the gates of Appendix E.

Paired comparisons. Pairing seeds by index on the held-out split, the local system's margin over the narrow-learner night is +3.1 points (bootstrap 95% interval [+1.4, +5.4], six seed pairs), over the same-learner night +2.6 ([+0.2, +5.2], six pairs; the night's seed variance makes this one fragile, and on the second split it shrinks to +0.3), over unmasked interleaved replay +6.3 ([+3.8, +9.2]), over BP+ER +2.8 ([+2.5, +3.2]), over the silent mask +3.6 ([+3.1, +4.1]), over rotation alone +0.1 ([−0.3, +0.5]), over DER++ +0.1 ([−0.3, +0.4]), over ER-ACE +1.5 ([+1.1, +1.9]) and over A-GEM +23.2 ([+18.3, +28.6]).

Updates and wall-clock. The headline schedule makes 16,885 masked updates against the night's 480 (35×; counts from the reported runs, whose stream is nine tenths of the training set). Replay cost is reported in samples (R = replay batches × batch size). The night costs $480 \times 256 \approx 1.2 \times 10^5$ samples per run; the headline configuration (one 16-sample micro-batch per waking batch) costs $16{,}885 \times 16 \approx 2.7 \times 10^5$ (2.2× the night) for 91.6±0.3. Two schedules meet the night's budget within 1.1×: eight-sample micro-batches at every waking batch (1.35×10^5, 91.1 ± 0.4) and 16-sample micro-batches at every second one (8,442 updates, 1.35×10^5, 91.2 ± 0.2 sequential, 93.1 ± 0.2 static); pressure-triggered bursts of 16 at a third of the events reach 88.6 ± 1.6 (Appendix E). Training wall-clock on one CPU thread is 8.8 min for the headline run against 1.4 for the night, 1.0 with no replay and 8.6 for unmasked interleaved replay (one run each, same protocol and thread count).

D Further ablations: optimiser state, replay gain, rotation gates

Optimiser state. Masked Adam and heavy-ball SGD tie on the held-out split ($91.5 \pm 0.4/94.1 \pm 0.1$ against $91.6 \pm 0.3/94.1 \pm 0.2$, sequential/static), and letting Adam's moments advance outside the mask adds nothing ($91.6 \pm 0.6/94.0 \pm 0.3$); we keep the single velocity as the least optimiser state that stays confined to the mask, not for accuracy. With momentum zero the default replay step (3η per micro-batch against the batch-scaled waking step $\eta\, m/256$) diverges or starves hidden activity at every learning rate tested; the waking path alone learns normally at the matched effective step, so the velocity's averaging of the 16-sample replay micro-batches is what heavy-ball provides. Re-scaling the replay gain by hand makes pure SGD viable at $88.4 \pm 0.5/89.4 \pm 0.3$, still 3.2/4.7 points behind.

Replay gain. Sweeping the replay gain over $\{1/16, 1/4, 1, 3, 10\}$ (default 3; $\eta_r = 3\eta$ against a waking step of $\eta/16$) moves masked replay $67.2 \pm 3.0 \to 86.8 \pm 1.1 \to 90.8 \pm 0.1 \to 91.6 \pm 0.3 \to 74.2 \pm 32.1$ and unmasked replay without rotation $61.6 \pm 7.2 \to 81.4 \pm 4.5 \to 85.7 \pm 2.2 \to 85.4 \pm 3.5 \to 47.3 \pm 30.7$ (sequential; three seeds, six at gain 10): masked replay peaks at the default, unmasked replay plateaus from gain 1 to 3, gain 10 destabilises both (two of six seeds collapse for the masked system, four for the unmasked), and no replay gain makes the unmasked control competitive.

Rotation gates. At the 10% activity fraction the relative-novelty gate of Appendix E, committed to bouts of $M{=}2048$ batches, gives $91.8 \pm 0.3/94.4 \pm 0.3$ against the always-on default's $91.6/94.1$, within seed noise; the same gate evaluated per batch gives $87.5 \pm 1.6/94.7 \pm 0.1$ because its flicker destroys the heavy-ball velocity's averaging. Rotation's static price is 1.3 points at this fraction (94.1 against 95.4 for the same mask without rotation); from 10 to 25% the sequential axis is flat while the static axis rises (development-phase sweep, official test split).

Exactness on other configurations. The rates in Section 3 are population rates over every replay step of a full run under the reported protocol. In development runs (official test split, full training set), the hidden-code rate was 0.32% for the default configuration and 0.46% on raw CIFAR-10 (one seed), where the prediction rate rises to 3.6% because low-margin predictions flip more easily through the plastic readout.

E Sparse replay budgets: internal triggers and the novelty gate

The headline system replays at every batch; the mechanisms here schedule sparser budgets. *When to replay:* each unit integrates its own use, $S_i \leftarrow S_i + \bar{a}_i(X_t)$ with $\bar{a}_i(X_t)$ the fraction of the batch's samples on which it fired (a unit-level Process S [Borbély, 1982]), and a burst of n_b replay micro-batches fires when the mean pressure over the currently asleep units of all hidden layers exceeds θ; consolidation resets the pressure of the units that received it. *When to rotate:* with fast and slow exponential averages of the batch-mean squared output error e_t, $\bar{e}_f$ ($\alpha_f{=}0.1$) and $\bar{e}_s$ ($\alpha_s{=}0.002$), rotation runs while $\bar{e}_f \geq \beta\, \bar{e}_s$ ($\beta \approx 1.1$–1.5) or while $\bar{e}_f \geq \gamma e_0$ with e_0 the chance-level error of the first 20 batches ($\gamma \approx 0.5$): the stream is new relative to the learner's own recent history, or is not yet mastered. The detector is a standard drift monitor [Ross et al., 2012], loosely inspired by cholinergic novelty signalling [Hasselmo, 2006]; what is new is what it gates. A single absolute threshold cannot serve across the regimes considered, because a converged ten-class i.i.d. error exceeds any level a two-class task dips under. The gate is committed to minimum-duration bouts, loosely modelled on the consolidated bouts of the sleep–wake switch [Saper et al., 2005].

Figure 3 compares triggers at sparse budgets on split-MNIST (held-out split). At 12.5% of the replay events a clocked burst of 16 batches every 128 reaches 80.5 ± 6.1, no better than an even trickle at the same budget (78.5 ± 2.6); unit-level pressure triggering keeps 88.6 ± 1.6 at a third of the events; surprise triggering fires no event (18.8), because surprise arrives only after a task switch. This is the opposite of the offline night, whose best trigger in our sweeps is novelty-timed.

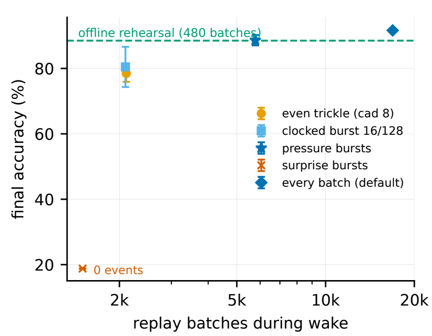

Figure 3: **Replay timing at sparse budgets:** accuracy against the number of waking replay events; trickles and clocked bursts fail alike, pressure-triggered bursts (star) keep most of the accuracy at a third of the events, surprise triggering fires no events in this regime.

F Isolation across replay batch sizes

Figure 4 extends Figure 2 to two-sample replay micro-batches and shows, per batch size, the seed-paired margin of the full system over rotation with unmasked replay (held-out split; six seeds at batches 2, 4 and 16, three at 8). From 16 down to 4 the margin is 0.1 $[-0.3, +0.5]$, 0.3 $[-0.2, +0.6]$ and 0.3 $[-0.5, +1.2]$: isolation costs nothing and buys nothing measurable in mean accuracy while the rotation keeps the live coalition tolerant of unmasked replay. At batch 2 the replay gradient is too noisy for either system. The isolated one falls to 79.0 ± 3.2 (forgetting 21.5 ± 4.2 against 6.4 at batch 16), with every seed between 75.5 and 84.3; the unmasked one bifurcates, with three seeds finishing at 82.5–84.8, slightly above their isolated counterparts ($+0.3$, $+1.8$, $+8.1$), one at 58.2 and two at chance (9.9). Leaking replay onto the live coalition is therefore not a pure loss, since the surviving seeds show that the extra plasticity can help, but it exposes a severe-failure mode (final accuracy below 60%) that the isolated system did not show in six seeds. Six seeds without a severe failure bound its frequency but do not prove the mode absent, and the masked system itself destabilises at replay gain 10 (Appendix D). The paired mean at batch 2 ($+24$ $[-0.2, +48.5]$) describes the unmasked failures, not a margin, and is kept out of the ranges quoted in the main text.

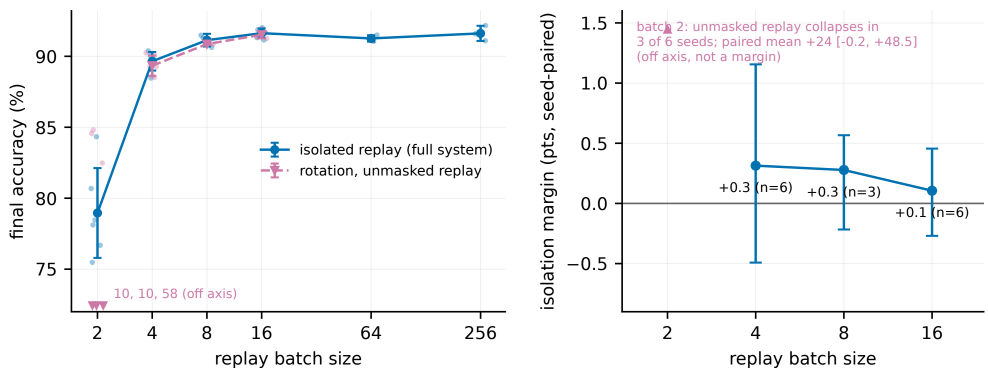

Figure 4: **Isolation across replay batch sizes** (held-out split). **Left:** split-MNIST accuracy against replay batch size for isolated replay (the full system) and for rotation with unmasked replay, every seed shown; at batch 2 no unmasked mean is drawn because three of six seeds fail severely (values at the axis floor). **Right:** the seed-paired margin of isolation with a 95% bootstrap interval at the batch sizes where the unmasked run does not collapse.

G A drive-referenced decay controller and depth

Every result in the main text runs the rule with a decay λ tuned once per dataset (10^{-3} on the image benchmarks). Split CIFAR-100 exposes what that constant hides: each class receives a tenth of the supervision events while the decay acts on every synapse at every step, so at the MNIST-tuned value the hidden weights follow the pure-decay trajectory and the network lands at chance (2.4% i.i.d. against 20–23% on a ten-class subset, development-phase numbers). We replace the constant by a per-layer ratio controller,

$$\lambda_\ell = \min\Big(\lambda_{\max},\ \rho\,\mathrm{EMA}\big[u^\ell\big]/\big(\overline{|W^\ell|} + \epsilon\big)\Big), \qquad \rho = 0.25,\ \lambda_{\max} = 0.02, \tag{5}$$

where u^ℓ is the mean magnitude of the post-optimiser, *pre-decay* waking increment of layer ℓ (the decay never feeds its own drive; EMA coefficient 0.02 over unmasked waking steps, $\epsilon = 10^{-12}$), and W^ℓ, $B^{\ell-1}$ share λ_ℓ: the decay's mean magnitude $\lambda_\ell\overline{|W^\ell|}$ is held at the fraction ρ of the mean data-driven update. ρ is the only control target; $\lambda_{\max}$, the EMA coefficient and ϵ are global safeguards never tuned per dataset. Isolation and rotation decide *where* and *when* to consolidate, the controller *how strongly*. The drive estimate must see only unamplified waking updates: inflated targets and the offline night's $3\times$-gain batches push the EMA to its clip and erase day learning.

Table 5: **One dataset-independent ratio target replaces per-dataset decay tuning.** Sequential / static accuracy under the tuned fixed decay (10^{-3}; 10^{-4} with $3\times$ targets as the CIFAR-100 rescue) and the controller (5) at $\rho = 0.25$ everywhere. Full system in every cell, held-out split, three seeds (six on split-MNIST and in the fixed-decay split CIFAR-10 cell); feature CIFAR feeds the images through a frozen random-patch front-end (1024-D); CIFAR-100 numbers are floor-regime and reported for ordering only (BP+ER 6.8 raw, 10.1 with features).

	fixed λ (tuned)		controller	
	seq.	static	seq.	static
split-MNIST	91.6 ± 0.3	94.1 ± 0.2	90.6 ± 0.6	93.5 ± 4.3
split CIFAR-10	28.4 ± 0.8	28.0 ± 1.2	26.2 ± 1.1	30.1 ± 0.2
feature CIFAR-10	42.3 ± 0.6	42.4 ± 1.1	39.1 ± 0.4	43.6 ± 0.6
split CIFAR-100	1.2 ± 0.1	1.0 ± 0.1	3.2 ± 0.8	4.1 ± 0.6
+ features	1.1 ± 0.3	1.1 ± 0.2	5.2 ± 1.6	6.4 ± 0.6

With the same ρ on every dataset (Table 5) the controller concedes 1–3 sequential points to the per-dataset tuned value and gains 1.2–5.3 static points on CIFAR-10, feature CIFAR-10 and both CIFAR-100 variants. On MNIST it gains 1.1 on the second held-out split (95.2 ± 0.4) but one of six seeds collapsed on the first (93.5 ± 4.3), so we claim no static gain there. On CIFAR-100 it lifts the system 2.7–4.7× over the manual rescue. The realised λ_ℓ spans 10^{-5} (MNIST streams) to 3×10^{-3} (raw CIFAR), so the per-dataset ladder was a signal-to-decay ratio in disguise.

Depth. Figure 5 varies the hidden depth. Under the tuned fixed decay without skips the static axis dies at four hidden layers ($10.1 \pm 0.2\%$, chance) and the sequential axis collapses at five (26.3 ± 20.2); under the controller every cell stays alive ($90.7 \to 88.9 \to 88.1$ sequential, $95.3 \to 94.8 \to 89.6$ static). The residual toll is architectural. ResNet-style skip synapses restore the thick error path: for $\ell \geq 3$ each deep hidden layer receives a bypass, $z^\ell = W^\ell a^{\ell-1} + S^\ell a^{\ell-2} + b^\ell$, with S^ℓ under the same Dale projection and a feedback twin $B_S^\ell \in \mathbb{R}^{n_{\ell+2} \times n_\ell}$ that carries the bypass target's burst back to its source, learned by the local product of the main text with the target layer's decay λ_ℓ and replay-masked by the same rule two layers apart, $M_{S,ij}^\ell = \mathbf{1}[\, j \notin \mathcal{A}^{\ell-2}(X) \ \vee \ i \notin \mathcal{A}^\ell(X) \,]$, so that Propositions 1 and 2 extend unchanged. The five-hidden-layer system then reaches $91.0 \pm 0.3/95.8 \pm 0.3$ (six seeds) against the two-layer default's $91.6/94.1$ and the two-layer controller's $90.6/93.5$. The bypass alone also rescues the fixed decay ($90.4/93.5$ at depth five), so error-path attenuation is a principal cause of the collapse (the fixed decay was not re-tuned per depth). BP+ER at the same widths is depth-insensitive ($88.9 \to 87.9$) but forgets 2× more.

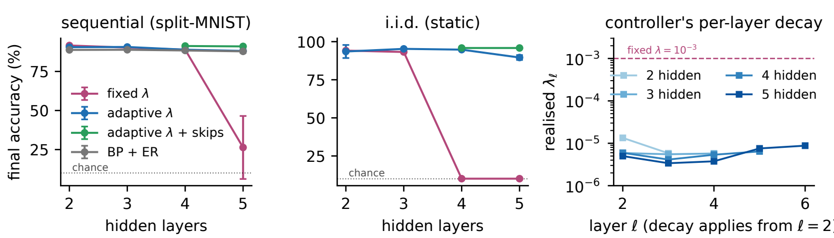

Figure 5: **Depth.** Sequential (left) and static (middle) accuracy against hidden depth (three seeds; six at the skip system's depth five): the fixed decay without skips collapses, the controller degrades gracefully, controller + skips hold the two-layer level. Right: the controller's realised per-layer decay.

H Development-phase numbers and a second held-out split

Configurations were selected on the official test splits (the development set) and frozen; the comparisons in the main text re-train them on nine tenths of the training data and evaluate on the held-out tenth, which took part in no selection. Table 6 places the development-set numbers beside the held-out ones and beside a second held-out tenth for the headline rows (drawn with a different seed; the two tenths share 9% of their samples, so they are not disjoint test sets). Over the 87 sequential configurations run under both protocols, reported numbers sit 1.1 points below the development numbers on average and rank-correlate with them at Spearman $\rho = 0.97$ (a robustness check, not a proof that selection bias is absent); every ordering carried by the paper holds on both held-out splits, with one magnitude that does not: the same-learner night, bimodal on the first split (89.0 ± 3.5), reaches 91.5 ± 0.5 on the second and trails the full system by 0.3 there. The adaptive decay's static margin on MNIST is likewise split-dependent (93.5 ± 4.3 with one collapsed seed on the first split, 95.2 ± 0.4 on the second). Table 7 gives the per-task accuracy matrix of the headline configuration on the reported split.

Table 6: **Development-set (official test split) numbers beside the held-out numbers and a second held-out split.** Sequential accuracy, with the static axis on its own row where the paper reports it; six seeds on the reported split, three on the development set and on the second split. Lower blocks: single pass, and raw split CIFAR-10.

	development set	held-out split 1	held-out split 2
isolated replay + rotation (default)	92.7 ± 0.3	91.6 ± 0.3	91.8 ± 0.4
static	94.7 ± 0.3	94.1 ± 0.2	94.1 ± 0.6
bout-committed gate	92.3 ± 0.2	91.8 ± 0.3	92.0 ± 0.6
rotation alone (unmasked replay)	92.1 ± 0.5	91.5 ± 0.3	91.8 ± 0.6
static	94.7 ± 0.1	93.7 ± 0.3	93.7 ± 0.3
silent mask (no rotation)	88.8 ± 0.8	88.0 ± 0.7	88.6 ± 1.3
unmasked interleaved replay	87.0 ± 1.1	85.4 ± 3.5	80.0 ± 14.1
BP + ER	89.3 ± 0.8	88.8 ± 0.3	88.5 ± 0.7
offline night, same learner	90.9 ± 3.2	89.0 ± 3.5	91.5 ± 0.5
offline night, narrow learner	90.7 ± 0.4	88.5 ± 2.8	89.3 ± 0.6
adaptive λ (drive)	91.7 ± 0.2	90.6 ± 0.6	90.8 ± 0.1
static	96.1 ± 0.2	93.5 ± 4.3	95.2 ± 0.4
five hidden layers + skips + adaptive λ	91.5 ± 0.4	91.0 ± 0.3	91.1 ± 0.3
static	96.2 ± 0.3	95.8 ± 0.3	95.7 ± 0.1
no buffer	19.6 ± 0.0	19.4 ± 0.0	19.5 ± 0.0
random rotation	90.8 ± 1.3	91.1 ± 0.7	90.7 ± 0.5
replay confined to the readout	27.3 ± 15.1	41.7 ± 10.9	38.1 ± 4.9
mirror isolation	77.9 ± 1.6	79.4 ± 3.5	79.2 ± 2.4
soft rotation	90.2 ± 0.4	89.9 ± 0.7	89.5 ± 0.4
BP + DER++	92.2 ± 0.5	91.5 ± 0.5	91.3 ± 0.3
BP + ER-ACE	90.4 ± 0.3	90.2 ± 0.4	90.4 ± 0.2
BP + A-GEM	73.6 ± 4.2	68.5 ± 7.0	70.4 ± 4.1
BP + k-WTA, interleaved replay (same schedule)	92.4 ± 0.4	91.7 ± 0.4	—
BP + k-WTA, isolated replay	84.3 ± 2.5	84.7 ± 2.6	—
BP + k-WTA, interleaved replay + rotation	92.9 ± 0.4	92.2 ± 0.3	—
BP + k-WTA, isolated replay + rotation	92.9 ± 0.2	92.1 ± 0.3	—
single pass (one epoch per task)			
isolated replay + rotation	91.8 ± 0.4	91.8 ± 0.3	91.8 ± 0.1
unmasked interleaved replay	68.3 ± 19.1	83.0 ± 14.9	83.9 ± 6.0
offline night, same learner	75.4 ± 0.5	76.9 ± 3.4	75.2 ± 5.5
BP + ER	90.1 ± 0.7	88.6 ± 0.2	88.2 ± 0.9
BP + DER++	—	90.1 ± 0.7	89.4 ± 0.3
split CIFAR-10 (raw)			
isolated replay + rotation	29.5 ± 0.8	28.4 ± 0.8	29.7 ± 0.9
offline night	26.2 ± 0.6	25.1 ± 2.7	26.5 ± 0.4
BP + ER	25.1 ± 1.5	24.6 ± 0.6	25.8 ± 0.8
BP + DER++	31.1 ± 0.3	30.7 ± 0.8	31.0 ± 0.6
BP + ER-ACE	31.7 ± 0.8	31.6 ± 1.0	31.7 ± 0.8
unmasked ER	17.0 ± 6.1	14.6 ± 4.6	15.5 ± 4.5
no buffer	16.4 ± 0.2	16.3 ± 0.1	16.6 ± 0.0

Table 7: **Per-task accuracy matrix** of the headline configuration on split-MNIST (held-out split, mean over six seeds): row = after training task i, column = accuracy on task j. Forgetting F, the mean of the diagonal minus the last row over tasks 1–4, is 6.4 ± 0.6; the drop is concentrated on tasks 2–3 and the first task is retained.

after task	T1	T2	T3	T4	T5
1	99.8				
2	98.7	97.2			
3	97.7	92.7	97.0		
4	97.5	89.6	92.9	97.5	
5	96.7	88.1	89.7	91.5	91.8